BTS–VII(R/S)–11–24–2961 Reg. No. | | | | | | | | |

B

B.Tech. Degree VII Semester Regular/Supplementary Examination November 2024

ME 19-205-0703 MACHINE DESIGN II

(2019 Scheme)

Time: 3 Hours Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Understand the customers need, formulate the problem and draw the design specifications.
CO2: Understand component behavior subjected to loads and identify the failure criteria.
CO3: Apply the concepts of stress analysis, theories of failure and material science to analyze, design and/or select commonly used machine components.
CO4: Analyze the stresses and strains induced in a machine element.
CO5: Design machine components using theories of failure.
CO6: Get basic knowledge of the preparation of working drawings by including the manufacturing details like tolerance, surface finish etc.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 –Analyze, L5 – Evaluate, L6 – Create
PO – Programme Outcome

(Use of Design Data Handbook is permitted)
(Data not given may be suitably assumed. All such assumptions must be clearly stated)

PART A

(Answer *ALL* questions)

		$(8 \times 3 = 24)$	Marks	BL	CO	PO
I.	(a)	Define positive and friction clutches. Explain the advantages of positive clutches over friction clutches.	3	L1	2	1
	(b)	List important factors upon which the selection of belt drive depends.	3	L1	2	1
	(c)	Define following terms used in gears. (i) Circular pitch. (ii) Pressure angle. (iii) Gear ratio.	3	L2	1	1,2
	(d)	Define overlap in helical gears and explain how helical angle effect end thrust. Also define formative number of teeth in helical gears.	3	L2	2	1,2
	(e)	Explain the theory of hydrodynamic lubrication with neat sketch.	3	L2	2	1
	(f)	A ball bearing is subjected to a radial force of 2500 N and an axial force of 1000 N. The dynamic load carrying capacity of the bearing is 7350 N. The values of X and Y factors are 0.56 and 1.6 respectively. The shaft is rotating at 720 rpm. Calculate the life of the bearing.	3	L3	1	1
	(g)	How can you represent surface finish and tolerance in working drawings?	3	L2	6	1,2
	(h)	What are the general design recommendations for casting?	3	L2	6	1,2

PART B

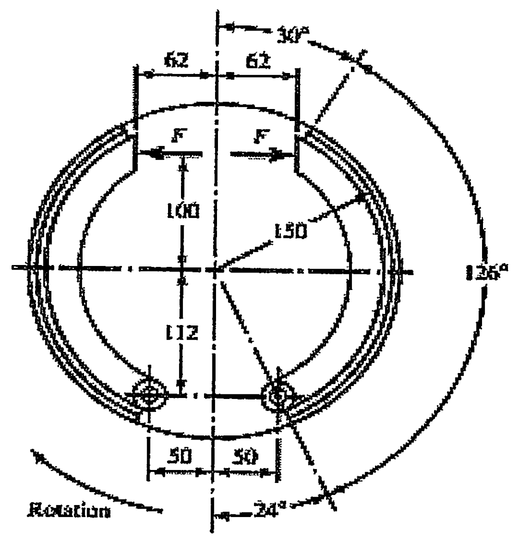
		$(4 \times 12 = 48)$				
II.	(a)	A multiple disc clutch, steel on bronze, is to transmit 4.5 kW at 750 rpm. The inner radius of the contact is 40 mm and outer radius of the contact is 70 mm. The clutch operates in oil with an expected coefficient of friction of 0.1. The average allowable pressure is 0.35 N/mm^2. Find (i) the total number of steel and bronze discs. (ii) the actual axial force required. (iii) the actual average pressure. (iv) the actual maximum pressure.	8	L3	3	1,2,3

(P.T.O.)

		Question	Marks	BL	CO	PO
	(b)	An engine developing 45 kW at 1000 rpm is fitted with a cone clutch built inside the flywheel. The cone has a face angle of 12.5° and a maximum mean diameter of 500 mm. The coefficient of friction is 0.2. The normal pressure on the clutch face is not to exceed 0.1 N/mm^2. Determine (i) the face width required. (ii) the axial spring force necessary to engage the clutch.	4	L3	3	1,2,3

OR

		Question	Marks	BL	CO	PO
III.	(a)	The brake shown in the figure is 300 mm in diameter and is actuated by a mechanism that exerts the same force F on each shoe. The shoes are identical and have a face width of 32 mm. The lining is molded asbestos having a coefficient of friction of 0.32 and a pressure limitation of 1000 kPa. Estimate the maximum (i) actuating force F. (ii) braking capacity. (iii) hinge-pin reactions.	8	L3	3	1,2,3



		Question	Marks	BL	CO	PO
	(b)	A leather belt 9 mm × 250 mm is used to drive a cast iron pulley 900 mm in diameter at 336 rpm. If the active arc on the smaller pulley is 120° and the stress in tight side is 2 MPa, find the power capacity of the belt. The density of leather may be taken as 980 kg/m^3 and the coefficient of friction of leather on iron is 0.35.	4	L3	3	1,2,3
IV.	(a)	A bronze spur pinion rotating at 600 r.p.m. drives a cast iron spur gear at a transmission ratio of 4: 1. The allowable static stresses for the bronze pinion and cast iron gear are 84 MPa and 105 MPa respectively. The pinion has 16 standard 20° full depth involute teeth of module 8 mm. The face width of both the gears is 90 mm. Find the power that can be transmitted from the standpoint of strength.	5	L3	4	1,2,3
	(b)	A helical cast steel gear with 30° helix angle has to transmit 35 kW at 2000 r.p.m. If the gear has 25 teeth, find the necessary module, pitch diameters and face width for 20° full depth involute teeth. The static stress for cast steel may be taken as100 MPa. The face width may be taken as 3 times the normal pitch. The tooth form factor is given by the expression $y' = 0.154 - 0.912/Z_e$, where Z_e represents the equivalent number of teeth. The velocity factor is given by $C_v = 6/(6 + v)$, Where v is the peripheral speed of the gear in m/s.	7	L3	4	1,2,3

OR

(Continued)

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			Marks	BL	CO	PO
V.	(a)	A pair of cast iron bevel gears connects two shafts at right angles. The pitch diameters of the pinion and gear are 80 mm and 100 mm respectively. The tooth profiles of the gears are of 14½° composite form. The allowable static stress for both the gears is 55 MPa. If the pinion transmits 2.75 kW at 1100 r.p.m., find the module and number of teeth on each gear from the standpoint of strength and check the design from the standpoint of wear. Take surface endurance limit as 630 MPa and modulus of elasticity for cast iron as 84 kN/mm^2.	8	L3	4	1,2,3
	(b)	A triple threaded worm has teeth of 6 mm module and pitch circle diameter of 50 mm. If the worm gear has 30 teeth of pressure angle 14½° and the coefficient of friction of the worm gearing is 0.05, find (i) the lead angle of the worm. (ii) velocity ratio. (iii) centre distance. (iv) efficiency of the worm gearing.	4	L3	4	1,2,3
VI.		The load on a journal bearing is 150 kN due to turbine shaft of 300 mm diameter running at 1800 rpm. Determine the following: (i) Length of the bearing if the allowable bearing pressure 1.6 N/mm^2. (ii) Amount of heat to be removed by the lubricant per minute, if the bearing temperature is 60°C, and the viscosity of oil at 60°C is 20 centipoise (0.02 kg/m-s) and the bearing clearance is 0.25 mm. (iii) Sommerfeld number.	12	L3	5	1,2,3
		OR				
VII.	(a)	A ball bearing supporting a bevel gear shaft is subjected to a radial load of 4000 N and a thrust load of 5000 N. Find dynamic load rating and equivalent load on a bearing when operating at a speed of 1600 rpm for an average life of 5 years at 10 hours per day.	4	L3	5	1,2,3
	(b)	A ball bearing is required to resist a radial load of 10 kN and a thrust load of 5 kN. The average life of the bearing is to be 5000 hours with inner race rotation at 980 rpm. What basic dynamic load rating must be used in selecting the bearing? If the bearing is to have a life of 5000 hours at a reliability of 97%, what is the required basic dynamic load rating?	8	L3	5	1,2,3
VIII.	(a)	Explain design considerations for forging.	8	L2	6	1,2,3
	(b)	Explain design recommendations for machined or turned parts.	4	L2	6	1,2,3
		OR				
IX.	(a)	Describe the design considerations for welded parts.	8	L2	6	1,2,3
	(b)	What are the considerations for design for manufacturing and assembly (DFMA)?	4	L2	6	1,2,3

Bloom's Taxonomy Level's
L1 – 5%, L2 – 12.50%, L3 – 82.5%.

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